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Dementia classification using MR imaging and clinical data with voting based machine learning models

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Abstract

Dementia is one of the leading causes of severe cognitive decline, it induces memory loss and impairs the daily life of millions of people worldwide. In this work, we consider the classification of dementia using magnetic resonance (MR) imaging and clinical data with machine learning models. We adapt univariate feature selection in the MR data pre-processing step as a filter-based feature selection. Bagged decision trees are also implemented to estimate the important features for achieving good classification accuracy. Several ensemble learning-based machine learning approaches, namely gradient boosting (GB), extreme gradient boost (XGB), voting-based, and random forest (RF) classifiers, are considered for the diagnosis of dementia. Moreover, we propose voting-based classifiers that train on an ensemble of numerous basic machine learning models, such as the extra trees classifier, RF, GB, and XGB. The implementation of a voting-based approach is one of the important contributions, and the performance of different classifiers are evaluated in terms of precision, accuracy, recall, and F1 score. Moreover, the receiver operating characteristic curve (ROC) and area under the ROC curve (AUC) are used as metrics for comparing these classifiers. Experimental results show that the voting-based classifiers

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often perform better compared to the RF, GB, and XGB in terms of precision, recall, and accuracy, thereby indicating the promise of differentiating dementia from imaging and clinical data.

Keywords Dementia classification · MR imaging · Random forest · XGB classifier · Voting classifiers · Gradient boosting classifier · Feature selection

1 Introduction

Alzheimer's disease (AD) is very common and it is one of the more familiar forms of senile dementia, affecting more than three million people in the USA alone. Though mainly occurs at the later stages of age 60 and beyond, some symptoms of AD can also manifest early in the range of 30–50 years [5, 31]. Dementia can affect people's cognitive abilities, thereby placing a burden on individual care and the healthcare system. There is still no efficacious medical treatment to slow down the progression of dementia. Therefore, there is an acute need to focus on timely intervention, at the early stages, and if possible, to mitigate or delay the progressive disease. The clinical diagnosis is dependent on the comprehensive medical history given by patients and their families, along with neuropsychological tests and neurological examinations. Other tests, including imaging modalities such as magnetic resonance (MR), computed tomography (CT), and hematology-based measurements, can be augmented with physician diagnosis to rule out further causes of dementia. Neuropsychological tests play an important role in detecting dysfunctions in human "cognitive domains". Although there have been various clinical measures for the early diagnosis of dementia, numerous subjectivities still exist [5, 23, 31]. Therefore, it is of great significance to improve better diagnostic tools for dementia with machine learning models on available patient imaging and clinical data.

Due to dementia, a patient's memory can be lost gradually, and it creates obstacles to perform even daily activities in severe cases. Humans perform several repeated daily activities by remembering and retrieving information from the cerebral hemisphere (memory containing lob in the brain). Dementia mainly causes damages in the process of retrieving data from the brain and causes the brain cell connections to degenerate and die, thereby destroying memory and other important mental functions. An important senile dementia type, especially related to memory loss and confusion in patients, is Alzheimer's disease (AD). However, one of the most severe forms of dementia, as noted by World Health Organization (WHO) [62], is vascular dementia that can cause a stroke. If people suffer from dementia before 65 years of age, then this type of dementia can be denoted as early-onset dementia. If people suffer from dementia before 65 years of age, then this type of dementia can be denoted as late-onset dementia. According to the World Alzheimer Report of the year 2015, the global number of people suffering from dementia is estimated to be 46.8 million. There will be an increase of 74.7 million estimated by 2030 and 131.5 million by 2050 [64]. Further, based on the World Alzheimer Report of the year 2019, the attitude towards dementia indicates that dementia patients can be hidden from outside, especially in South East Asia (16.3%) and the African (7.1%) regions of the world [61]. The following salient points are important motivating factors in developing and advocating automatic detection and classification of AD:

- Alzheimer's can cause severe problems with memory. As a result, patients can face many obstacles to performing their normal daily activities.

- Alzheimer has several symptoms that are curable, however, once the 'Alzheimer' is recognized it is not fully curable except for management to improve symptoms temporarily. Moreover, there are several identifiable pre-symptoms of AD.
- Recent research indicates that there is no specific age range when AD can occur. Therefore, there is no certain time frame to start the diagnosis and move to care [19].
- In North America, 'Alzheimer' is the sixth leading factor in overall mortality. Increasingly, developing countries also have a considerable population of AD patients requiring urgent classification of stages.

Table 1 indicates the four important stages of AD, namely (i) pre-diagnosis, (ii) primary, (iii) middle, and (iv) final stages. Imaging and especially magnetic resonance imaging (MRI) based features can predict the rate of decline of AD patients and may help guide therapy and care. For reaching that stage doctors and researchers will have to make proper use of automatic data-driven machine learning methods. These methods can be able to predict the progress of a patient from mild cognitive impairment to dementia accurately [63, 3, 12, 14, 25, 29, 33, 40, 54, 58, 59, 66].

In this work, we propose novel approaches combining imaging data with machine learning models for classifying dementia. Our main contributions are in proposing a hybrid classification via voting approaches, wherein we considered two cases, namely hard and soft voting. Despite the combination looking complex, we note that the classifiers are explainable, and we show by feature scoring, and feature importance for the data at hand. In comparison with random forest (RF), gradient boosting (GB), logistic regression (LR), extreme gradient boosting (XGB), support vector machine (SVM), our proposed voting-based method obtains better precision, recall, and F1 scores. We also show, by extensive experiments, that our

Table 1 Stages of Alzheimer's disease [12, 33, 54, 58, 59]

Stage	Descriptions
Pre-diagnosis	1) Identify the problems 2) Make an appointment with the professionals 3) Take the advice 4) Meet with the professionals after certain days to check whether it is case or not
Primary Stage	1) As it is an early stage of the disorder, make sure the plans for the cure 2) Make a plan to overcome the worst situation as it will increase day by day 3) Accept the change in life, keep contact with the doctor, and keep constant with the family members 4) Should be familiar with the medical terms and make safety plans 5) Increase the connection to the Alzheimer community and attend the season regularly if possible
Middle Stage	1) As the longest stage, the patient should have a proper activity plan which includes the daily routine and meditation 2) Should learn to increase the behavior and communication 3) Should not leave alone and make a proper plan for upcoming days with appropriate financial considerations 4) Should use modern tech items like GPS and other devices to keep in touch with medical professionals and relatives 5) Legal affairs should be brought under light
Final Stage	1) The patient should be provided 24×7 care 2) Should keep him/her in a jolly mood and provide proper medication 3) Contact the community and medical personal regularly 4) Make a chart for development and plan for daily activities 5) Make the space for the patient the keep the connection with the family members

approach obtains consistently higher accuracy and area under the curve (AUC) values. Further, compared to other techniques [9, 10, 28, 46, 65, 67], with similar outcomes available in the literature, our voting-based classifiers obtain higher accuracy and AUC.

The rest of the paper is organized as follows. Section 2 provides a literature review on dementia where several related works are presented. Section 3 describes the data used for testing different machine learning models for dementia classification. Section 4 presents the methodology details. Section 5 describes experimental results and analysis, feature selection, and ranking techniques along with the classification using an ensemble learning-based approach. We also provide comparisons with existing machine learning classifiers for accuracy and ROC curves. Finally, Section 6 provides the concluding remarks.

2 Literature reviews

Dementia is a disease of the brain that attacks mental activities i.e., thought, rationality, and memory. It is mainly caused either by traumatic injuries or old age, with around 60–70% of cases attributed to AD [37]. There are typically three processes of the AD diagnosis: (i) consultation with a physician, (ii) perform required neuropsychological tests, (iii) additional imaging, and especially MRI scans [44]. Some researchers propose to study the diagnosis of dementia wherein they use neuropsychological measures such as Consortium to Establish a Registry for Alzheimer's Disease (CERAD) [44], Mini-Mental State Examination (MMSE) [22]. A number of machine and deep learning algorithms used on several medical data [2, 43]. As a result of artificial intelligence (AI) [41, 42], huge amounts of data are being combined with pieces from many digital sources, including electronic health records, medical imaging processing, and cancer data [20, 21, 47]. Farid et al. [27] proposed a composite hybrid feature selection (CHFS) method in open access series of imaging studies (OASIS) Alzheimer's medical datasets for extracting new features and choosing the best features for a clinical diagnosis of AD. The selection of appropriate features can lead to good classification results with a high amount of recall and accuracy. Farid et al. [27] applied the stacked classification method, along with six basic models such as support vector machine (SVM), J48, Naïve Bayes, multilayer perceptron (MLP), JRIP, and random forest (RF) were used for stacking purposes. Shree and Sheshadri [53] adopted J48, Naïve Bayes, RF, and JRIP in order to detect and diagnose Alzheimer's patients. Among the four classifiers, they report that the highest accuracy was obtained with the Naïve Bayes classifier. The limitations of [53] were that they did not describe the feature selection technique and they did not pre-process the dataset used in the experiment. Proper feature solation is a crucial factor, and a pre-process of the dataset is a crucial factor for increasing the precision, recall, accuracy, etc. for different datasets. Their results are not verified for the different datasets. The MMSE scores and tests six domains of interest in the test subjects: (i) attention and calculation, (ii) registration, (iii) orientation, (iv) language, (v) recall, and (vi) constructional ability [39]. Battineni et al. [6] used SVM for predicting dementia from OASIS dataset and achieved precision and accuracy of 64.18% and 68.75%, respectively, for $C = 100$ where C is a regularization parameter [6]. However, the overall accuracy achieved in this work is comparatively low. Chen et al. [18] adopted some ML methods such as Naïve Bayes, Bayesian-network classifier, SVM, decision trees, and MLP in order to diagnose very mild dementia and compared those ML methods with logistic regression and discriminant analysis, and the dataset used consisted of 83 and 30 subjects for training and testing purpose. The authors of [57] use EEG data for classifying Alzheimer's and

normal cognition using SVM classifier. Therefore, EEG Epochs present 79.9% accuracy and the SVM provides 87% accuracy. The authors of [60] compared dementia screening and the Clinical Dementia Rating (CDR) scoring using Neural Network, Decision Tree, Naive Bayes, and SVM, and their results showed that Naive Bayes provides a more accurate result than SVM which obtained the lowest accuracy. Facal et al. [26] used some ML algorithms such as RF, SVM, LR, Gradient boosting, KNN, etc. to a dataset of 169 patients for predicting conversion and non-conversion to dementia. For tenfold cross-validation, the highest precision rate (88%) and lowest precision rate (56%) were achieved by using the Gradient boosting classifier and KNN classifier, respectively. In [51], the prediction of CDR was performed by applying seven machine learning algorithms such as Bayes Network, Naïve Bayes, Logistic Regression, Bagging, SVM, MLP, and RF. Some DL techniques were applied to biomedical data. For training deep neural networks, the authors of [1] showed how to build a pathologist-cured visual dataset along with data augmentation techniques for dramatically increasing dataset dimensionality, which is particularly important in deep CNN training due to problems with overfitting. Another work [52] suggested a ML based technique where MRI cortical thickness assisted in predicting the progression of Parkinson disease patients from moderate cognitive impairment to dementia at the individual level, with better results when combined with clinical factors. Apart from MRI, retinal vasculature was used for classifying Alzheimer's disease [56]. The pipeline was able to obtain 82.44% average classification accuracy using UK Biobank data. This work enhanced the interpretability of this pipeline by including a saliency analysis. Small vessels in retinal images offered more information for identifying Alzheimer's disorders, according to saliency analysis, which is in line with prior findings. Despite the importance of this work, this work provided a low accuracy [56]. Moreover, Herzog et al. [34] offered an ML based pipeline for classifying dementia. Numerous fields, including image analysis and processing, have adopted methods of deep learning. In most cases, no feature engineering or image preprocessing was required before classification, which added to their attractiveness. These approaches function well when dealing with biased and large datasets. This kind of deep neural network, dubbed a convolutional neural network (CNN), was built exclusively for image processing. It can distinguish AD from NC with a 98% accuracy rate and predict whether a patient would develop stable or transformed MCI with a 75% accuracy rate using just MRI data [4]. Numerous neural network models were introduced for the analysis of non-imaging medical data. The authors of [55] built three deep learning architectures for interpreting and processing genetic and clinical data using PET and MRI imaging data. Their algorithms detected dementia and cognitive impairment with an accuracy of 86%.

3 Data

The experimental dataset is provided by open access series of imaging studies (OASIS) (www.oasis-brains.org), which is available in Kaggle repository [36]. The dataset comprises a longitudinal collection of 150 subjects whose age range was 60–96. Each subject has been scanned on two or other visits, divided by at least one year and the total number of imaging sessions is 373. Three or four individual T1-weighted MRI scans attained in single scan sessions had been included for each subject and they were right-handed. There were men and women among the subjects. 72 peoples were categorized as non-dementia while 64 people were categorized as dementia patients during their initial visits and remained so for succeeding scans. Among the 64 people, 51 individuals suffered from mild to moderate Alzheimer's

disease. The remaining fourteen people were characterized as non-dementia during their initial visit and were later considered dementia patients at the time of their next visits. There is some clinical information as described below:

(a) EDUC – Education level.
(b) SES – Socio-Economic Status (SES).
(c) MMSE – Mini-Mental State Examination (MMSE) score with range: 0–30, and 0 is worst and 30 is best.
(d) CDR – Clinical Dementia Rating (CDR) is a 5-point scale used to characterize functional performance and six domains of cognition applicable to Alzheimer's disease and inter-related dementias. The sis domains are Memory, Orientation, Judgment and Problem Explaining, Community Affairs, Hobbies, and Home, and Personal Care. 0 indicates no dementia, 0.5 indicates very mild Alzheimer's disease (AD), 1 indicates mild AD, 2 indicates moderate AD.
(e) eTIV – estimated Total Intracranial Volume (eTIV) and it is in mm^3 units. Total intracranial volume (TIV/ICV) is a significant covariate for volumetric evaluations of the brain and brain regions, specifically in the study of neurodegenerative diseases by providing a proxy of maximum pre-morbid brain volume. Men showed an approximately ~12% larger eTIV than women.
(f) nWBV – normalized Whole-Brain Volume (nWBV), expressed as a percent of all voxels in the atlas-masked image that are labeled as gray or white matter by the automated tissue segmentation process.
(g) ASF – Atlas Scaling Factor (unitless). A computed scaling factor that transforms the native-space brain and skull to the atlas target i.e., the determinant of the transform matrix.

Table 2 illustrates the statistical summary of the dataset, and Fig. 1 illustrates how many male and female patients are included in the dementia versus non-dementia patients group respectively.

Figure 2 illustrates the box plots of SES, EDUC, MMSE, CDR, eTIV, ASF, and nWBV (from left to right), respectively, and the most accurate feature is eTIV. The box plots show the interquartile range (IQR) and are calculated by the upper quartile minus the lower quartile. This range is presented by the size of the box with the middle fifty percentages of scores shown.

Table 2 Summary of the dataset in terms of clinical information

Statistical Parameters	Visit	MR Delay	Age	EDUC	SES	MMSE	CDR	eTIV	nWBV	ASF
Mean	1.882	595.105	77.013	14.598	2.460	27.342	0.291	1488.129	0.729	1.195
Standard Deviation	0.923	635.485	7.641	2.876	1.134	3.683	0.375	176.139	0.037	0.138
Min	1.000	0.000	60.000	6.000	1.000	4.000	0.000	1106.000	0.644	0.876
25%	1.000	0.000	71.000	12.000	2.000	27.000	0.000	1357.000	0.700	1.099
50%	2.000	552.000	77.000	15.000	2.000	29.000	0.000	1470.000	0.729	1.194
75%	2.000	873.000	82.000	16.000	3.000	30.000	0.500	1597.000	0.756	1.293
Max	5.000	2639.000	98.000	23.000	5.000	30.000	2.000	2004.000	0.837	1.587

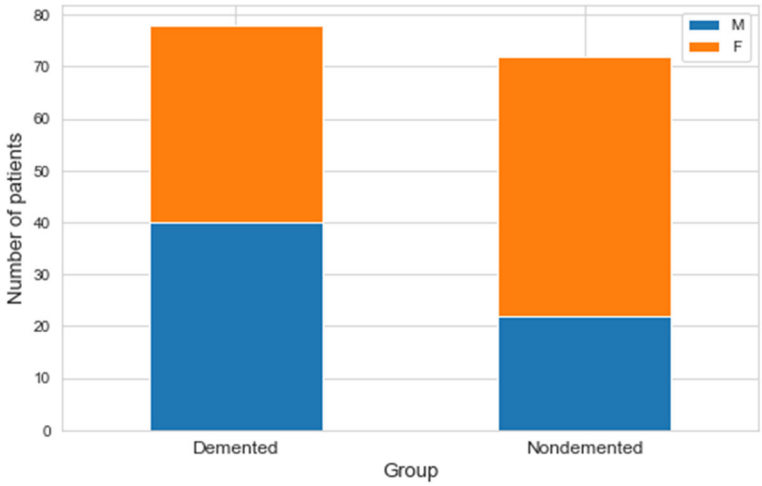

Fig. 1 Division of the number of dementia and non-dementia patients in terms of gender in the dataset

4 Methods

4.1 Our approach

The dataset used in this research is collected from a publicly available Kaggle repository [36]. Python-based scikit-learn machine learning library is used in testing different ML classifiers.

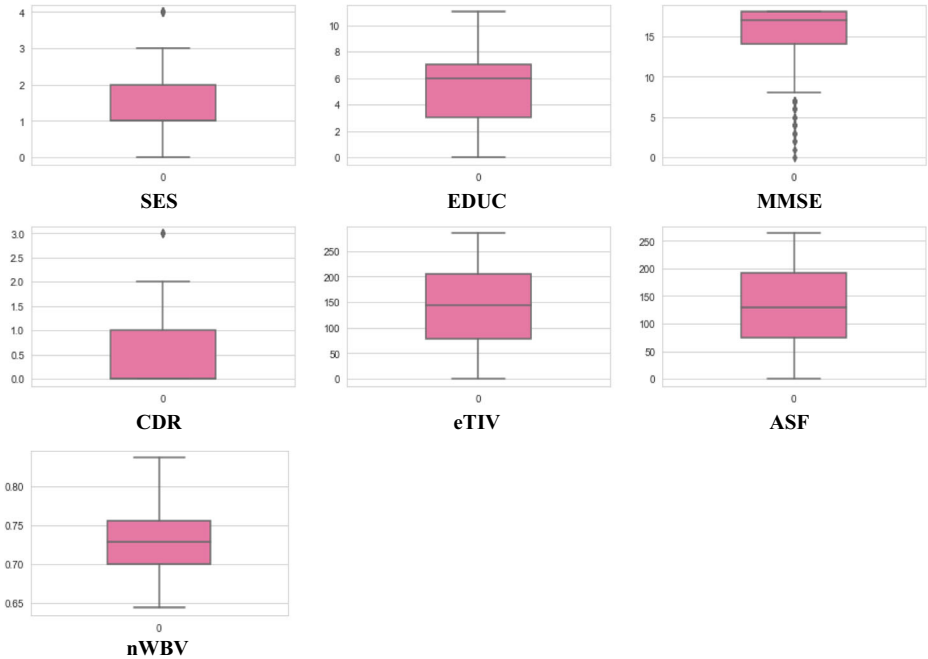

Fig. 2 Interquartile ranges of Socio-Economic Status (SES), Education (EDUC) level, Mini-Mental State Examination (MMSE) score, Clinical Dementia Rate (CDR), Estimated Total Intracranial Volume (eTIV), Atlas Scaling Factor (ASF), and normalized Whole-Brain Volume (nWBV). The top feature is eTIV

Scikit-learn is built upon pandas, NumPy, and Matplotlib, etc. The overall workflow diagram is given in Fig. 3. For classification, we consider the following machine learning methods: random forest (RF), gradient boosting (GB), and extreme gradient boosting (XGB). Moreover, we also propose a hybrid classification based on a voting-based approach. Three types of voting classifiers are considered in this work for hard and soft voting cases.

4.2 Feature selection and ranking

The scikit-learn library of Python is used for evaluating the experimental results [7, 8, 45]. The OASIS dataset is used for the experiments, which contain 373 samples, and a total of 15 attributes. The most important features (top 10 features) are selected by applying the univariate feature selection method. Univariate feature selection is a type of filtering-based method that is comparatively faster than other techniques. In this method, each feature or attribute is scored individually depending on some specified criteria. The important features are chosen based on the high scoring value. In scikit-learn, *SelectKBest()* function is used for selecting the best 10 features, see Table 3 that presents the result of feature ranking using the data-driven diagnosis of dementia patients.

4.3 Feature extraction

The PCA is one method of extracting features. Through the use of principle component analysis (PCA), a big dataset can be reduced in feature count or dimensionality by using a small number of primary features instead of many features. Using PCA, the original dataset is reduced to a smaller number of variables while retaining the vast majority of its original information. As a result, PCA decreases model complexity while simultaneously decreasing

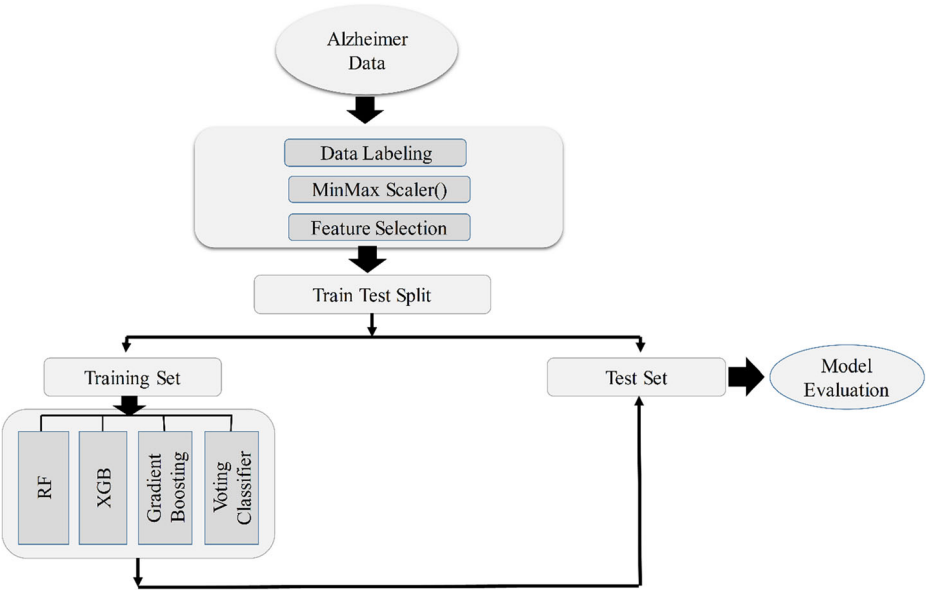

Fig. 3 The overall workflow is considered here with various classifiers for dementia classification

Table 3 Feature scoring using univariate selection

Feature	Score
MR Delay	18,753.039095
MRI ID	17,984.219265
eTIV	15,890.309345
Subject ID	7475.343058
nWBV	4524.003857
Age	931.230392
EDUC	330.059444
Visit	245.257497
SES	241.895211
MMSE	241.325693

the likelihood of overfitting. Because PCA is sensitive to data scaling, characteristics recorded on various scale factors must be harmonized. Prior to PCA, characteristics must be normalized such that the PCA procedure treats all features equally. In this study, we utilized *PCA()* function from scikit-learn to carry out the PCA analysis for extracting features.

4.4 Classification

In this section, we detail our hybrid classification via a voting-based approach. We consider two different types of voting, hard and soft. We use the classifiers such as the RF [13], GB [30], XGB [17], LR [35], and SVM [11] as ensemble combination within our voting-based approach.

Hard voting is a majority-based voting mechanism [24] that is usually applied in ensemble classification. Predict the diseases with the biggest number of votes from various models. Hard voting can be applied to increase the classification results [38]. Every one of these model characterization results is processed for each test instance of features. Majority-based results provide the final output of the prediction.

On the other hand, Soft voting is a method of ensemble some classifiers in which the output of the prediction is based on probabilities P for the various classifiers [15]. This method is feasibly suggested if the classifiers are highly constant.

It can be expressed mathematically as follows:

$$Class(x) = argmax_{c_i \in dom(y)} \left(\sum_k g(y_k(x), c_i) \right) \tag{1}$$

where $y_k(x)$ denotes the k-th classifier's classification and $g(y, c)$ denotes an indicator function expressed as:

$$g(y, c) = \begin{cases} 0, & y \neq c \\ 1, & y = c \end{cases} \tag{2}$$

and while probabilistic classifiers are used, we can calculate $y_k(x)$ (crisp classification) in the following way:

$$y_k(x) = argmax_{c_i \in dom(y)} \hat{P_{M_k}}(y = c_i | x) \tag{3}$$

$\widehat{P_{M_k}}(y = c_i|x)$ implies the probability that an instance x belongs to class c where each label y and each instance i. The value of *dom (y)* indicates the number of classes where the *arg max* function returns the index value that corresponds to the array's highest value.

Algorithm 1 shows the overall classification with voting and with three types of classifiers.

Algorithm 1. Dementia people detection using a voting classifier

Input: A List of Selected Features

Output: Classification Report, Confusion Matrix, ROC Curve

Steps:

1. StandardScaler() function is used to standardize the selected features.

2. Apply voting classifier combining {Extra Trees Classifier (), Gradient Boosting Classifier (), ensemble. Random Forest Classifier (criterion='gini', max_depth=7, max_features='auto', n_estimators=200), XGB Classifier ()} on the selected features by using the ensemble learning.

Voting 1: Gradient Boosting Classifier (), ensemble, Random Forest Classifier (criterion='gini', max_depth=7, max_features='auto', n_estimators=200), XGB Classifier ()

Voting 2: ensemble.Random Forest Classifier (criterion='gini', max_depth=7, max_features='auto', n_estimators=200), svm.SVC(probability=True)), XGB Classifier ()

Voting 3: Gradient Boosting Classifier (), Logistic Regression(tol=0.0001, C=1.0, fit_intercept=True, intercept_scaling=1,random_state=0, max_iter=50)), ensemble.Random Forest Classifier (criterion='gini', max_depth=7, max_features='auto', n_estimators=200))

3. Train the model using selected features where the random state is 10.

4. Predict result using the test dataset

5. Evaluate the accuracy of the classifier using accuracy_score () function

6. Use confusion_matrix () function for the evaluation of FP, FN, TP and TN.

7. Use classification_report () function to calculate recall value, precision value, and F_1-score.

5 Experimental results and discussion

5.1 Feature ranking

Table 3 shows the ranking of features based on the univariate feature selection algorithm. It can be seen that the top three significant attributes are MR Delay, MRI ID, and eTIV. Moreover, verification of feature importance for the most accurate decisions is obtained by using Bagged Decision Trees, see Table 4. The top feature is the eTIV, and the least important is the group attribute that represents dementia and non-dementia as 1 and 0 respectively. Figure 4 shows a correlation matrix for understanding these features further. As can be seen,

Table 4 Estimation of feature importance using bagged decision trees on the OASIS dataset

Feature	Score
eTIV	0.138
nWBV	0.113
MRI ID	0.112
Age	0.108
Subject ID	0.088
MR Delay	0.087
MMSE	0.085
Visit	0.077
EDUC	0.065
SES	0.059
CDR	0.038
Group	0.021

despite the higher scores of MRI ID and Subject ID (as shown in Table 3), these two are not important enough compared to other features.

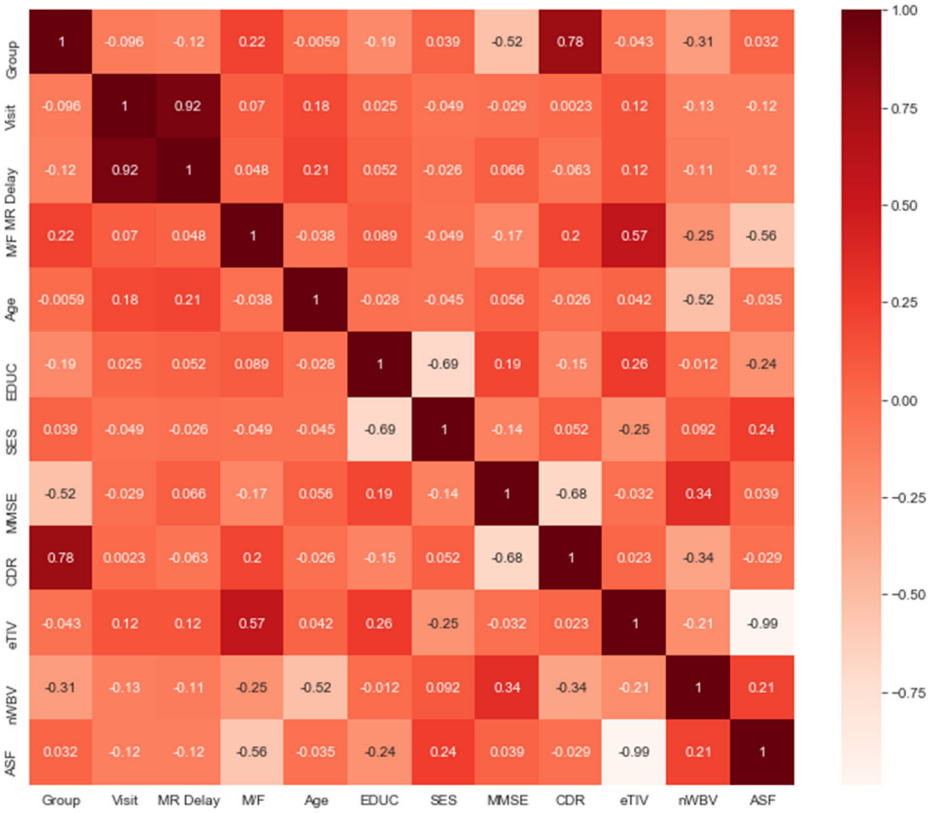

Fig. 4 Correlation matrix of features

5.2 Classification results

We next provide detailed experimental results with various classifiers along with different standard metrics such as precision, recall, accuracy, and F1 score. We also provide receiver operator curve (ROC) and area under the ROC (AUC) to show classifier performances. Table 5 describes the performance of several classifiers when testing data is at 15% and training data is at 85% along with the three voting approaches explained in Section 4.3. We have used four classifiers namely RF, XGB, voting, and GB. It can be seen from Table 5 that, voting 1 classifier shows the best precision, recall, F_1 score, and specificity values compared to others. The precision, recall, and F1-score for voting 1 (RF, GB, XGB) (Soft) classifier having only 8 features are 92%, 91%, and 91%, respectively. Weighted average precision and recall are the average scores of precision and recall obtained for both classes 0 and 1 since it is a binary classification. For example, the precision values obtained for 1 and 0 classes are 0.89 and 0.72, respectively. Then the weighted average precision will be 0.81.

Figure 5 presents ROC curves for four classifiers when testing data is 30% of the dataset along with their corresponding accuracy and AUC values. From Fig. 5, it can be seen that the Voting 1 (RF, GB, XGB) (Soft) classifier achieves the highest accuracy (91.17%) among them. The values of AUC of RF, XGB Classifier, Gradient Boosting classifier are 90%, 89%, and 91%, respectively. In contrast, the Soft Voting 1 Classifier has the highest AUC values compared to them. The values of AUC of Voting 1, Voting 2, Voting 3 classifiers (Soft) are 91%, 89%, and 88%, respectively.

Table 6 illustrates the performance of various classifiers when 30% of samples are used for testing and 70% of samples are used for training. It can be seen that the precision, recall, and F_1

Table 5 Comparison of performance of different classifiers when testing data is 15% in terms of weighted average precision, recall, F1-scores, and corresponding confusion matrices

Classifier	Weighted Average Precision	Weighted Average Recall	Weighted Average F1 Score	Confusion Matrix
RF	90%	89%	89%	[[26 1] [5 24]]
XGB	90%	89%	89%	[[25 2] [4 25]]
GB	91%	91%	91%	[[25 2] [3 26]]
Voting 1 (RF, GB, XGB) (Hard)	90%	89%	89%	[[25 2] [4 25]]
Voting 1 (RF, GB, XGB) (Soft)	92%	91%	91%	[[26 1] [4 25]]
Voting 2 (RF, LR, GB) (Hard)	88%	88%	87%	[[25 2] [5 24]]
Voting 2 (RF, LR, GB) (Soft)	90%	89%	89%	[[25 2] [4 25]]
Voting 3 (RF, SVM XGB) (Hard)	88%	88%	87%	[[25 2] [5 24]]
Voting 3 (RF, SVM XGB) (Soft)	88%	88%	87%	[[25 2] [5 24]]

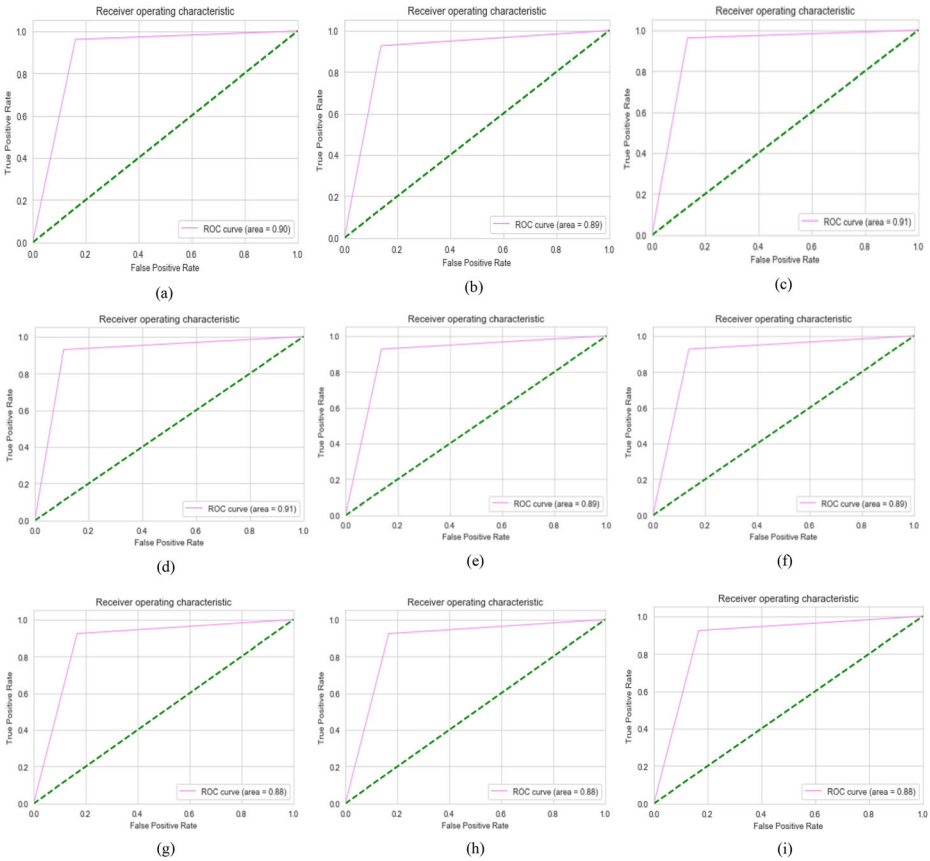

Fig. 5 Visualization of ROC curves of **a** RF - Accuracy = 89.29%, AUC = 90%, **b** XGB - Accuracy = 89.39%, AUC = 89%, **c** GB - Accuracy = 91.02%, AUC = 91%, **d** Voting 1 (RF, GB, XGB) (soft) - Accuracy = 91.17%, AUC = 91%, **e** Voting 1 (RF, GB, XGB) (hard), − Accuracy = 89.29%, AUC = 89%, **f** Voting 2 (RF, LR, GB) (soft) - Accuracy = 89.29%, AUC = 89%, **g** Voting 2 (RF, LR, GB) (hard) - Accuracy = 87.50%, AUC = 88%, **h** Voting 3 (RF, SVM, XGB) (hard) - Accuracy = 87.50%, AUC = 88%, **i** Voting 3 (RF, SVM, XGB) (soft) - Accuracy = 87.50%, AUC = 88%, when testing data is 15% for the purpose of comparison

score for Voting 2 classifier (Soft) are calculated as 88%, 87%, and 87%, respectively. Note that these values for voting classifiers are better than other compared classifiers.

Figure 6 presents ROC curves for four classifiers when testing data is 30% of the dataset along with their corresponding accuracy and AUC values. From Fig. 6, it can be seen that the accuracy values obtained by RF, XGB, GB classifiers are 87.5%, 82.14%, and 83.93%, respectively. The accuracy obtained by Voting 1, Voting 2, and Voting 3 (Soft) are 87.71%, 86.61%, and 84.92%, respectively. So, the Voting classifier, in general, achieved the highest accuracy among them. RF, XGB, GB classifiers obtained similar accuracy values.

Figure 7 shows the bar chart for comparative analysis of various classifiers along with our proposed voting-based approach with 15% testing and 85% training data in terms of AUC, accuracy, and recall. Figure 8 shows the bar chart for comparative analysis of various

Table 6 Comparison of performance of different classifiers when testing data is 30% in terms of weighted average precision, recall, F1-scores, and corresponding confusion matrices

Classifier	Weighted Average Precision	Weighted Average Recall	Weighted Average F_1 score	Confusion Matrix
RF	88%	88%	87%	[[49 3] [11 49]]
XGB	83%	82%	82%	[[45 7] [13 47]]
GB	84%	84%	84%	[[46 6] [12 48]]
Voting 1 (RF, GB, XGB) (Hard)	86%	85%	85%	[[47 5] [12 48]]
Voting 1 (RF, GB, XGB) (Soft)	87%	86%	86%	[[49 3] [13 47]]
Voting 2 (RF, LR, GB) (Hard)	86%	85%	85%	[[49 3] [14 46]]
Voting 2 (RF, LR, GB) (Soft)	88%	87%	87%	[[49 3] [12 48]]
Voting 3 (RF, SVM, XGB) (Hard)	85%	84%	84%	[[47 5] [13 47]]
Voting 3 (RF, SVM XGB) (Soft)	84%	84%	84%	[[46 6] [12 48]]

classifiers along with our proposed voting-based approach with 30% testing and 70% training data in terms of AUC, accuracy, and recall. As can be seen, the Voting 1 (Soft) approach obtains overall good AUC, accuracy, and recall values when compared to other classifiers. Moreover, when using lower training data (70%), the Voting (Soft) obtained better accuracy than other voting and other classifiers, indicating the robustness of the voting-based ensemble learning model. Gradient boosting performs equally well at the higher training data (85%), however, at the lower training data (70%), its performance degrades considerably with the accuracy dropping from 91.02% to 83.93%.

Finally, in Table 7 we show a comparative summary of AUC, and accuracy values of the proposed voting classifier method compared with other research works. We also provide the corresponding validation method used in each of the compared models in terms of cross-validation (CV) or holdout. For the proposed voting classifier, the results are provided for the case holdout method. It is shown that the classification accuracy for voting is 91%, which is larger than other models (RF, GB, XGB, Voting 1, Voting 2, and Voting 3) implemented in this work for the holdout method, respectively. Overall, the performances of the proposed model are better than those reported for RF having 55.3% [65], SVM (radial basis function) having 86.7% [67], and LDS [46] having 74.74%, SVM having 69.15% [46]. Further comparison with other models [16, 32, 48, 50] also shows that our proposed approach consistently obtains higher AUC and accuracy values.

5.3 Discussion

Detection and classification of dementia is an important task in computer-aided diagnosis systems for neurogenerative disorders. By combining imaging data with clinical

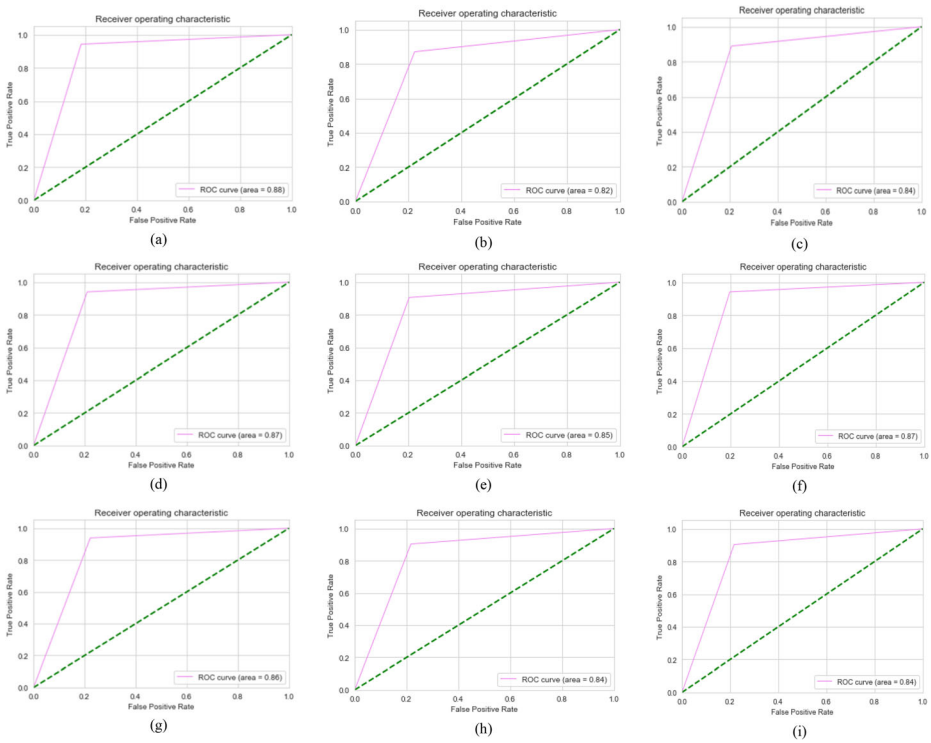

Fig. 6 Visualization of ROC curves of **a** RF - Accuracy = 87.50%, AUC = 88%, **b** XGB - Accuracy = 82.14%, AUC = 82%, **c** GB - Accuracy = 83.93%, AUC = 84%, **d** Voting 1 (RF, GB, XGB) (soft) - Accuracy = 88.71%, AUC = 87%, **e** Voting 1 (RF, GB, XGB) (hard), − Accuracy = 84.82%, AUC = 85%, **f** Voting 2 (RF, LR, GB) (soft) - Accuracy = 86.61%, AUC = 87%, **g** Voting 2 (RF, LR, GB) (hard) - Accuracy = 84.82%, AUC = 86%, **h** Voting 3 (RF, SVM, XGB) (hard) - Accuracy = 83.92%, AUC = 84%, **i** Voting 3 (RF, SVM, XGB) (soft) - Accuracy = 84.92%, AUC = 84%, when testing data is 30% for the purpose of comparison

variables, our proposed machine learning models-based approach can obtain accurate predictions overall. We used univariate feature selection and analysis shows that estimated Total Intracranial Volume (eTIV) is an important feature and it has been previously implicated in imaging-based studies [49]. We tested various machine learning classifiers in a voting-based ensemble approach with two variants – soft and hard. Our results indicate that soft voting 1 with RF, GB, and XGB obtained a weighted F1 score of 91% with only 8 features when the test data is 15% and training on 85% data. Similarly, soft voting 2 with RF, LR, GB obtained a weighted F1 score of 87% when the test data is 30% and the training on 70% data. Furthermore, precision, accuracy, and recall are favorable for the interpretable voting-based approaches, and also the performance matches or exceeds other related models from the literature [16, 28, 32, 46, 48, 50, 65, 67]. Moreover, classification from imaging as well as clinical variables data with machine learning classifier has proven to obtain better predictive results than with either one type of data alone. In the future, it will be interesting to fuse and apply other ensemble approaches for the differential diagnosis of dementia.

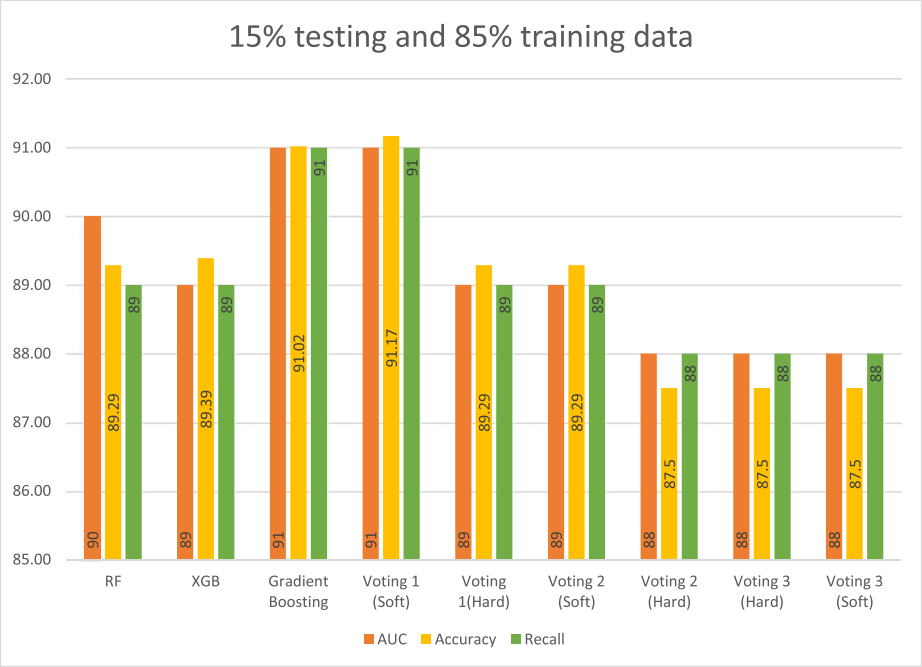

Fig. 7 Comparison of various machine learning classifiers along with voting-based approach with 15% testing and 85% training data in terms of AUC, Accuracy, and Recall

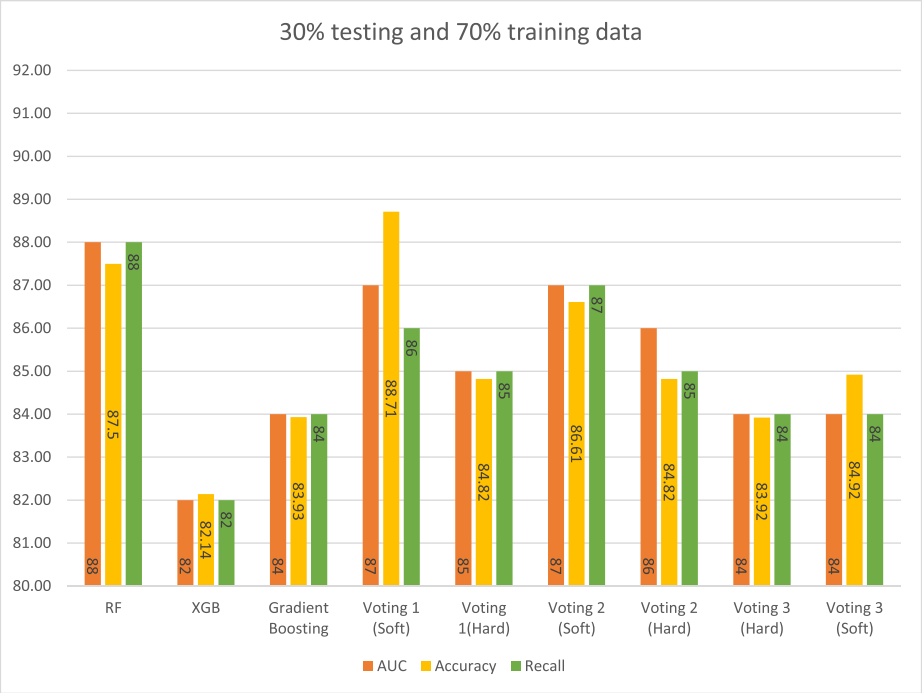

Fig. 8 Comparison of various machine learning classifiers along with voting-based approach with 30% testing and 70% training data in terms of AUC, Accuracy, and Recall

Table 7 Comparison with other classification models from the literature along with RF, GB, and our proposed voting-based approach

Sl. No.	Reference	Classification model	Validation Method	AUC	Accuracy
1	Ye et al. [65]	RF	CV	71%	55.3%
2	Filipovych et al. [28]	RF	CV	61%	–
4	Zhang et al. [67]	SVM (Radial Basis Function)	CV	–	86.7%
5	Zhang et al. [67]	SVM (Linear Kernel Function)	CV	–	91.47%
6	Moradi et al. [46]	LDS	CV	76.61%	74.74%
7	Moradi et al. [46]	SVM	CV	74.30%	69.15%
8	Castellazzi [16]	ANFIS	CV	85.27%	85.25%
9	Gill et al. [32]	Logistic model tree	CV	86%	84.4%
10	Park et al. [48]	RF	CV	68.3%	64.1%
11	Rohini et al. [50]	SVM	–	78%	89%
12	Tian et al. [56]	SVM	CV	–	82.44%
13	Proposed Approach	RF	Hold out	90%	89.29%
		XGB		89%	89.39%
		GB		91%	91.02%
		Voting 1 (RF, GB, XGB) (Soft)		91%	91.17%
		Voting 2 (RF, LR, GB) (Soft)		89%	89.29%
		Voting 2 (RF, LR, GB) (Hard)		88%	87.50%
		Voting 3 (RF, SVM, XGB) (Soft)		88%	87.50%
		Voting 3 (RF, SVM, XGB) (Hard)		88%	87.50%

6 Conclusion

A data-driven diagnosis of dementia is introduced by combining machine learning classifiers with a voting-based approach. For this, a dataset of 373 samples along with imaging data was used. After feature extraction, feature selection is executed to select the 10 most important features of the dataset. It is shown here that the most significant attribute is estimated Total Intracranial Volume (eTIV). The voting-based classifiers were considered in this work with the combination of some base classification models such as random forests, gradient boosting, extreme gradient boosting, support vector machines, etc. for classifying dementia or non-dementia. To separate the data samples into training and testing ones, the hold-out method is applied. Training and testing sets are prepared under different conditions, such as testing of 15%, or 30% samples of the dataset [36]. Results show that the voting classifier performs the best in the data-driven diagnosis of dementia. In the future, the effectiveness of extreme gradient boost, and gradient boosting classifiers, along with several other types of voting classifiers, should be evaluated for several different datasets.

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Declarations

Ethical approval The data used for this research work is collected from the publically available OASIS database and the authors do not violate the ethical statement.

Conflict of interest The authors declare that they have no conflict of interest.

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